Anand Nagaraj

Hardware Design Automation · Python · EDA Flows · AI-assisted RTL & Verification

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PROFILE

M.Sc. student in Design and Test of Integrated Circuits at TU Chemnitz with **3.5 years of industry experience** in digital IC design and verification and a strong focus on **automating design flows**: Python regression frameworks, LLM-assisted RTL and testbench generation, curated RTL datasets for AI models, and scripted RTL-to-GDSII flows. Seeking a **working student position or internship** in EDA, AI-driven design workflows or design automation.

PROFESSIONAL EXPERIENCE

RTL Design & Verification Engineer (founding team) 11/2024 – 02/2026
Semiconductor IP start-up Bengaluru, India

- Curated and verified **RTL datasets** (FSMs, AMBA AHB/AXI bridges, memory controllers) used to train AI/LLM models for automated RTL generation, linting and simulation.
- Built a **Python regression framework** that runs simulations, aggregates logs and tracks functional coverage; used AI-assisted scripting for testbench boilerplate and error-log clustering.
- Designed and verified NoC router RTL and custom RISC-V (Ibex) instruction extensions (SystemVerilog, UVM, cocotb, SVA).
- Guided junior engineers and interns through the full digital implementation flow to GDSII.

RTL Design & Verification Engineer 09/2022 – 10/2024
Hardware engineering services company Kochi / Bengaluru, India

- Co-developed **"ChipGPT"**, an automated RTL generation and verification workflow integrating Python and LLMs with EDA tools for linting, synthesis, simulation and static timing analysis.
- Built Python agent pipelines to parse simulation logs, cluster warnings and speed up regression triage.
- Designed and verified a PCIe Transaction Layer IP (Verilog, UVM, cocotb); synthesis and timing closure with Design Compiler; FPGA bring-up in Vivado.

KEY PROJECTS

Post-quantum TLS retrofit on a RISC-V board – NextSEMI 2026 hackathon
C · OpenSSL 3.5 · ML-KEM · ML-DSA

- Retrofitted a TLS 1.3 client on an embedded RISC-V (RV64GC) Linux board with hybrid key exchange (X25519 + ML-KEM-768) and hybrid authentication (ECDSA P-256 + ML-DSA-44).
- Offset the post-quantum overhead with RISC-V bit-manipulation (Zba/Zbb) builds, multithreaded signature verification and CPU-frequency tuning: handshake **31 % faster than the legacy baseline** (14.5 vs 21.0 ms, 1,000 runs, own test server).

Dual-port SRAM macro – scripted RTL-to-GDSII flow
Verilog · OpenLane 2 · OpenROAD · cocotb

- 16 × 8 dual-port SRAM taken through the full OpenLane 2 / OpenROAD flow on SkyWater 130 nm: DRC/LVS clean, STA sign-off.
- cocotb tests for write-masking, read latency and simultaneous access. *github.com/AnandNag003/tt_moola_dp_sram*

Moola-V (MR5) – RV32IM core, automated regression
SystemVerilog · Python · cocotb · RISCOF

- Designed a 5-stage pipelined RV32IM core; automated Python/cocotb regression in lock-step with the Spike ISS.
- **100 % compliance on all 49 RISCOF suites**, re-run automatically on every change.
  github.com/AnandNag003/moola_riscv_core

AMBA AHB-to-APB bridge
Verilog · SystemVerilog · UVM

- Cycle-accurate FSM translating pipelined AHB transfers into APB cycles; address decoding, HREADYOUT wait states, error mapping.
- UVM environment (AHB master / APB slave agents), constrained-random sequences and SVA – full coverage closure.

EDUCATION

M.Sc. Design and Test of Integrated Circuits 04/2026 – present
Technische Universität Chemnitz Chemnitz, Germany

- Modules: Verification of Digital Systems, Integrated Circuit Design (transistor level), Design of Heterogeneous Systems, Advanced IC Technology, Test and Design of Mixed-Signal Systems.
- **Deutschlandstipendium** 2026/27 – national merit scholarship (industry-sponsored).

B.E. Electronics & Telecommunication Engineering 08/2018 – 04/2022
Visvesvaraya Technological University (BMSIT) Bengaluru, India

- CGPA 8.51 / 10 (German grade ≈ 1.8). **Best Outgoing Student** of the department; state merit scholarship (100 % tuition waiver).
- Modules: CMOS VLSI Design, Computer Architecture, Digital System Design, Microprocessors.

Advanced VLSI Design & Verification – 6-month full-time programme 04/2022 – 09/2022
VLSI training institute Bengaluru, India

- SystemVerilog, UVM testbench architecture, SVA, constrained-random verification, coverage closure, AMBA AXI / AHB / APB, QuestaSim.

TECHNICAL SKILLS

Automation	Python (OOP, regression frameworks, log parsing, coverage extraction), Bash, Tcl basics, Linux, Git
AI for hardware	LLM-assisted RTL / testbench generation, agent pipelines for EDA log triage, RTL dataset curation for model training
Digital design flow	RTL (SystemVerilog, Verilog, VHDL), synthesis (Design Compiler), STA, OpenLane 2 / OpenROAD RTL-to-GDSII, SkyWater 130 nm
Verification	UVM, SVA, cocotb, pyUVM, Spike ISS, RISCOF; QuestaSim, Xcelium, VCS, Verilator
Architectures	RISC-V RV32IM, Network-on-Chip, PCIe Transaction Layer, AMBA, SRAM
Research	IEEE conference publication; experimental benchmarking and result analysis

PUBLICATION

A. Nagaraj et al., “Peripheral Shared Memory Over PCI Express Link,” in *Proc. 7th IEEE International Conference on Innovative Data Communication Technologies and Application (ICIDCA)*, 2025. – Shared-memory throughput, buffer sizing and latency over PCIe.

HONOURS & SELECTED ACTIVITIES

- **NextSEMI 2026 semiconductor hackathon** (SEMICON Europa × TUM Venture Labs, Munich, 11/2026) – one of 25 participants selected across Europe; post-quantum TLS project (see Key Projects).
- **Deutschlandstipendium** 2026/27, TU Chemnitz.
- **Best Outgoing Student** 2022 and state merit scholarship (100 % tuition waiver), VTU / BMSIT.

LANGUAGES

English – fluent (C1) | German – A2 (B1 course in progress) | Kannada – native | Hindi – fluent

FURTHER ACTIVITIES & INTERESTS

Volunteer, IEEE Student Branch TU Chemnitz (technical workshops, semiconductor seminars). Interests: distance running, cricket, hiking in Saxony.

Chemnitz, November 2026